

PROFESSOR VELAZQUEZ

# INTRODUCTION TO MICROECONOMICS

FALL 2025



01 | Module

# Overview of Economic Thought and Modeling

**Pre-Work**

Read Chp 1 - 2 & Syllabus

**Agenda Day 1:**

5 min - Do Now

15 min - Logistics

20 min - Guided Notes

30 min - Activity

**WHAT YOU'LL NEED Activity**

pencil, paper

**Big Questions**

(1) What are the basic principles of economic thought and analysis?

(2) How do economist model consumer and producer tradeoffs under scarcity?

**Do Now**

There is one Hershey Kiss for everyone in this class to share? What is the best system to decide how to distribute this single piece of chocolate?

**Section I: Economic Thought**

**Part 1:** Understand how to use classroom resources in order to meet course expectations.

**Part 2:** Identify and explain three key principles in economic thought and their philosophical origins.

**Part 3:** Explore key features of economics as an academic discipline, and develop a human history timeline.

**Section II: Economic Modeling**

**Part 4:** Develop the the budget constraint to understand how economist model constrained decision-making – or decision making under scarcity.

**Part 5:** Develop the Production Possibilities Model to form a deeper understanding of opportunity cost and economic modeling.

**Part 6:** Use the PPM of two countries to explore comparative advantage conceptually and explain how trade is beneficial to growth and production.



# Economic Thought

01

| Section



# Course Overview

01 | Part



# PERSONAL NARRATIVE

My academic and professional areas of focus are personal finance, public policy and community-based organizing.

**What are your areas of interest?**

I grew up in South Jersey about 40 min from here! My mom still lives in South Jersey and my dad lives in Sabana Grande, Puerto Rico. I currently live with my partner Brandon at Princeton University.

2020 – graduate Saint Joseph's University, AB Economics

2020 - Started Teach for America in Philadelphia at Boy's Latin Middle School

2022 – University of Pennsylvania, MSED in Urban Education

I taught Special Education and Math in West Philadelphia for 3 year. After teaching, I started a new career in personal finance. I worked as a tax specialist for H&R Block and then as a financial specialist for ECS, a community based non-profit focused on ending intergenerational poverty for BIPOC, were I currently work.



# Economic Research

02



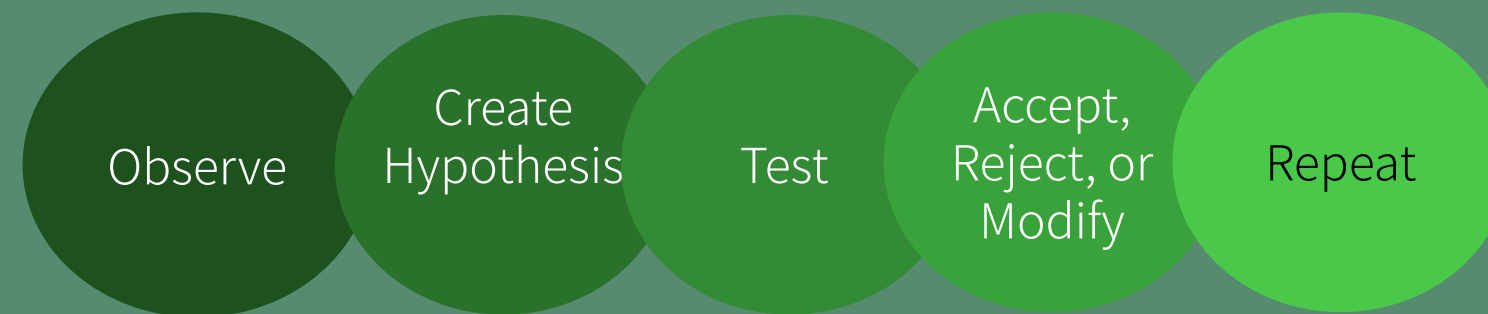
Part



# How do we know about economics?

## EMPIRICAL RESEARCH & THE SCIENTIFIC METHOD

**01** Using the scientific method, economists make statistically significant generalizations about our world (or test economic hypotheses) through natural experiments, lab experiments, data engineering and data analysis.



**Ceteris Paribus** (other things remaining the same)

With statistical programming, and their knowledge of theoretical mathematics, economists attempt to “isolate variables,” or observe a phenomenon, so that nothing but the **THING OF INTEREST** is changing.

## OUR HUMAN HISTORY

**02** Like other disciplines, economics is grounded in well-tested principles and shaped by both scientific reasoning and human history. Though relatively young compared to our long past, economics has developed into a vital field whose tools and frameworks now serve as the foundation for understanding an increasingly quantitative world.

**Did you know?** A Ph.D. in economics is a philosophy degree focused on quantitative social sciences.





# What is economic research?

## POSITIVE AND NORMATIVE STATEMENTS

01

### **POSITIVE** what is

- Economic statements that attempt to capture a phenomenon occurring at or over a specific point in time, must be based in fact or clear cause and effect relationships.
  - In 1933, the US unemployment rate rose to 25%.

### **NORMATIVE** what ought to be

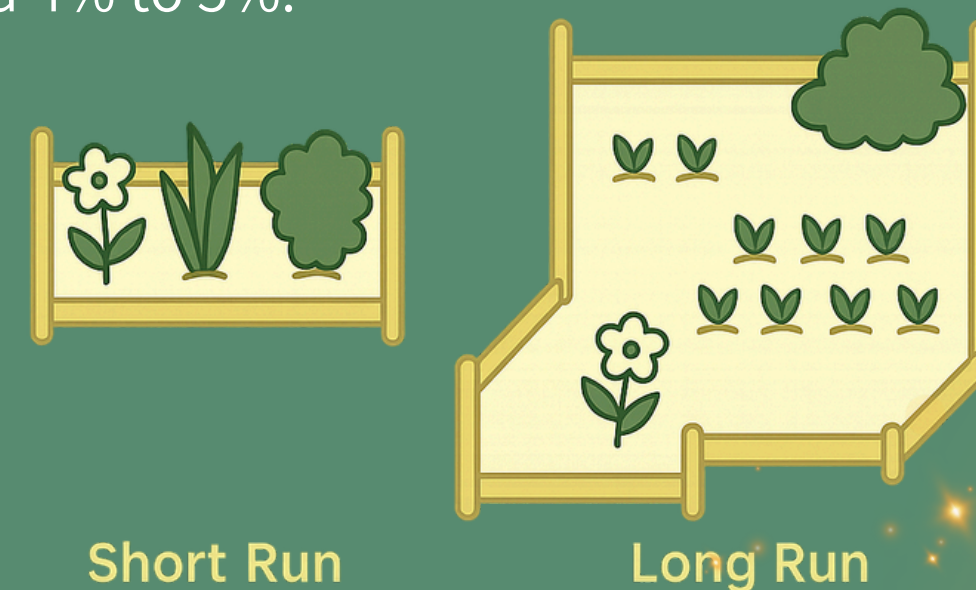
- Incorporate value judgments of how things should be based on fact and cause and effect relationships.
  - In the US economy, a desirable unemployment rate is around 4% to 5%.

## ANALYTICAL TIME-HORIZONS

02

**SHORT-RUN** All accessible resources are fixed (limited)

**LONG-RUN** All accessible resources are variable (changing)



## MICRO V. MACRO

03

**Micro** decision making by individual customers, workers, households, and businesses (a smaller system or speck within the glitter) “thinking about how a part of the game works”

**Macro** focused on economic growth, the business cycle, interest rates, inflation and government (the entire system of glitter) “thinking about the whole game”



# Economic Theory

02



Part

# What are general assumptions in neoclassical economics?

## SCARCITY, CHOICE & OPPORTUNITY COST IN THE WORLD...

01

The resources that are used in human production (land, labor, capital\*, and innovation) are **scarce resources** – limited supply and access. Humans are **capable of infinite desires** and have access to limited resources to meet them (time, raw materials, technology, mental capacity, etc.)

This becomes our **ECONOMIZING PROBLEM!!**

02

As a result of this, humans experience not only a scarcity of resource, but additionally a **scarcity of choice**. This means that for every choice made, there is a non-choice or **opportunity cost**.

If I use \$2 to buy a chocolate bar I cannot also use that same \$2 to buy a diet coke.

03

## LEAD TO RATIONAL, INCENTIVIZABLE & MARGINAL BEHAVIOR

Many economists assume that economic behavior is often **rational, self-interested** and driven by **incentives in the form of money**. Additionally, economists assume you can accurately quantify discrete parts of life (education, finance, public policy, etc.) and use this to inform purposeful, decision-making based on **mathematic analysis of change** or **marginal analysis**.

Economists like to call this, ‘**decision-making at the margin**’.

04

Yet, research across behavioral economics, experimental economics, and as far back as classical economics, shows that the relationship between incentives, choice, and financial decision-making is more complex.

While **money is a powerful motivator**, it operates alongside **social norms, identity, and intrinsic motivation**, particularly in contexts where **cooperation or altruism** are essential.



# Scarcity & Choice

Society's economizing problem – the factors of production are society's scarce resources.

## RESOURCES



**land**



**labor**



**capital**



**innovation**

## TERMS

**opportunity cost** - the non-choice, what you give up as a result of this scarcity

**capital** - man-made resources—such as tools, machines, buildings, and financial assets—used to produce goods and services. Unlike land or labor, capital must be created and invested in, and it increases productivity over time.

**scarcity of the mind** - **Mullainathan & Shafir (2013)** – show that scarcity, of money, time, or resources, captures attention and narrows focus. This “tunneling effect” reduces mental bandwidth, making it harder to plan for the long term and often leading to choices that seem irrational but are rational responses to pressure ← focus of my full-time job at ECS.

**The big question – What is the best system?**

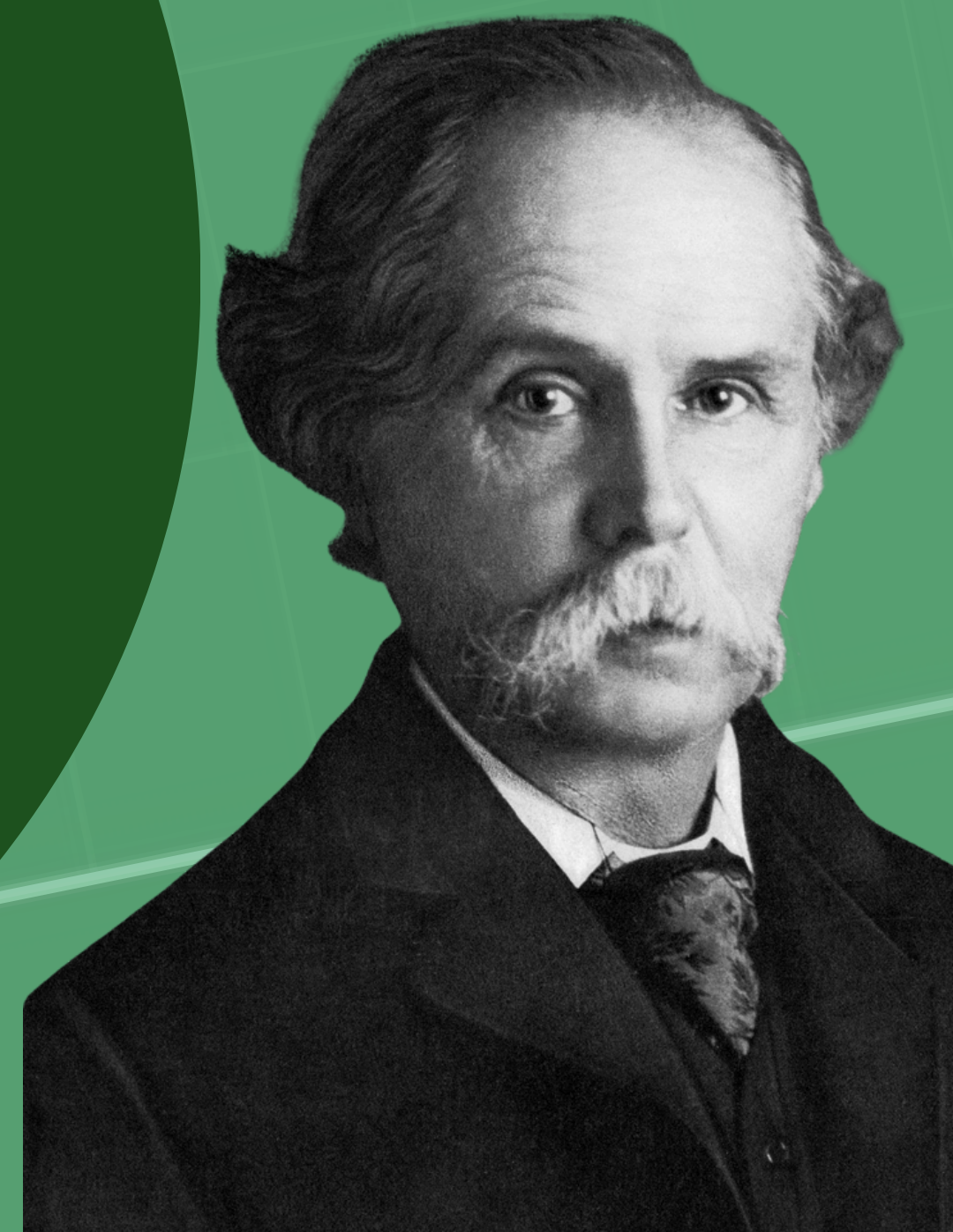




## RATIONAL THOUGHT

While the history of economic thought has ancient roots, many mark **1776** as the start of classical economics and western economic thought. This is because of the release of **Adam Smith's, "Wealth of Nations"** – which laid the groundwork for capitalist economic systems!

Following the development of **calculus** by Isaac Newton, the Marginal Revolution of the late 19th century — led by economists like Jevons, Menger, and Walras — formalized these earlier ideas. This culminated in the birth of **neoclassical economics** in 1890, when Alfred Marshall published *Principles of Economics*, a foundational text that continues to shape the way economics is used to shape your life today.





# Rational Thought

the power of being a “rational” being

## CLASSICAL & NEOCLASSICAL THEORIST

### **1776. Adam Smith – The Wealth of Nations**

→ Self-interest & “invisible hand” guide markets; division of labor drives growth.

### **1789. Jeremy Bentham – An Introduction to the Principles of Morals and Legislation**

→ Utilitarianism: individuals act to maximize pleasure, minimize pain.

### **1798. Thomas Malthus – Essays on the Principle of Population**

→ Population grows faster than food supply, creating scarcity pressures.

### **1817. David Ricardo – Principles of Political Economy and Taxation**

→ Theory of comparative advantage; distribution of income between land, labor, and capital.

### **1803. Jean-Baptiste Say – Treatise on Political Economy**

→ Say’s Law: supply creates its own demand.

### **1861. John Stuart Mill – Utilitarianism**

→ Refined utilitarianism: distinguished higher vs. lower pleasures; social welfare matters.

### **1871. William Stanley Jevons –**

→ Marginal utility theory; helped launch the marginal revolution.

### **1871. Carl Menger – Principles of Economics**

→ subjectivist theory of value

### **1874. Leon Walras – Elements of Pure Economics**

→ General equilibrium theory; formalized markets mathematically.

### **1890. Alfred Marshall – Principles of Economics**

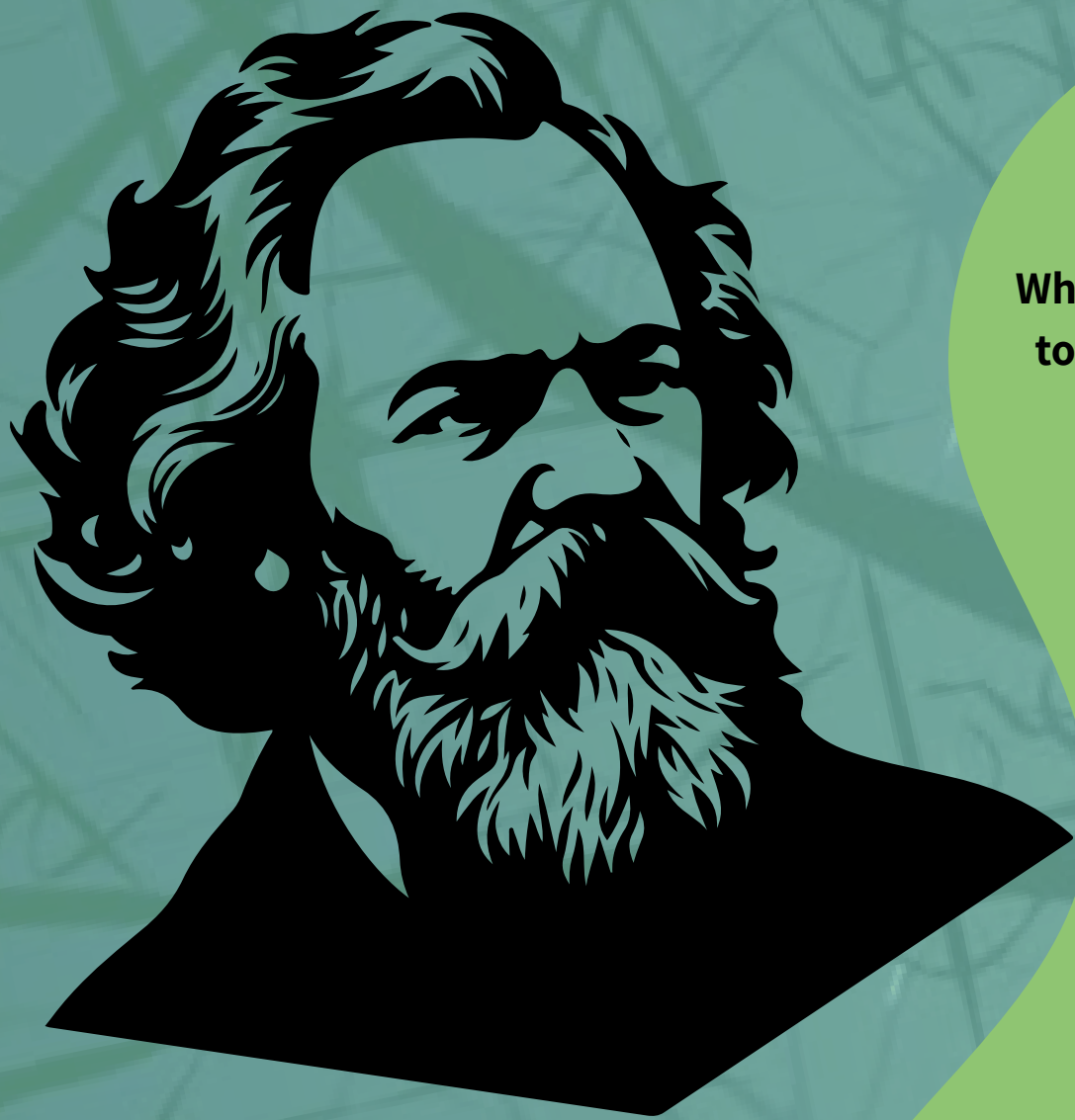
→ Formalized supply & demand, marginal utility, and equilibrium analysis.

### **1906. Vilfredo Pareto – Manual of Political Economy**

→ Pareto efficiency; distribution and welfare theory.

### **1976. Gary Becker – The Economic Approach to Human Behavior**

→ Applied rational choice to all human actions (crime, family, education).



# irRATIONAL THOUGHT

While these scholars are in no way “irrational” within their work, this term is used to help emphasis how these theorist challenge and extend neoclassical thought.

Karl Marx, in Das Kapital (1867), critiqued capitalism’s class structure and labor exploitation, offering a systemic alternative in the form of **command economies**.

Almost 150 years later using a similar historical approach to capital accumulation, **Thomas Piketty** uses historical data to show inequality as a systemic feature of capitalism in his book Capital in the 21<sup>st</sup> century.

In response to the Great Depression, **John Maynard Keynes** developed **macroeconomic theory**, emphasizing demand, uncertainty, and the need for government intervention.

**Behavioral economists** Daniel Kahneman and Amos Tversky challenged rational-agent models with Prospect Theory, highlighting cognitive biases in decision-making.

**Institutional economists** like Douglass North and Daron Acemoglu emphasized the role of laws, governance, and political structures in shaping economic outcomes.

In **development economics**, Esther Duflo, Abhijit Banerjee, and Michael Kremer advanced field experiments to fight poverty.





# irRational Thought

heterodox and emerging economics

## CRITIQUES & EXPANSIONS

### 1854. Wilhelm Roscher – Foundations of Political Economy

→ German Historical School: economics must reflect history, culture, and institutions.

### 1867. Karl Marx – Das Kapital, Volume I

→ Theory of surplus value; critique of exploitation under capitalism, important to understand **command economies**.

### 1899. Thorstein Veblen – The Theory of the Leisure Class

→ Introduced “conspicuous consumption”; behavior shaped by social status.

### 1947. Herbert Simon – Administrative Behavior

→ “Bounded rationality”: people satisfice rather than fully optimize decisions.

### 1970. George Akerlof – The Market for Lemons

→ Asymmetric information can cause market failures (e.g., used car markets).

### 1979. Daniel Kahneman & Amos Tversky – Prospect Theory

→ People value losses more than gains; challenged rational expected utility theory.

### 1990. Douglass North – Institutions, Institutional Change and Economic Performance

→ Institutions shape economic outcomes and long-term development.

### 1990. Elinor Ostrom – Governing the Commons

→ Communities can self-govern shared resources without state/market control.

### 2008. Richard Thaler & Cass Sunstein – Nudge

→ Choice architecture can improve decisions without restricting freedom.

### 2011. Esther Duflo\* & Abhijit Banerjee – Poor Economics

→ Poverty decisions are context-driven; RCTs reveal effective policy interventions.

### 2013. Thomas Piketty\* – Capital in the Twenty-First Century

→ Wealth grows faster than income ( $r > g$ ); capitalism naturally drives inequality.

# Building Marginal Thought

math is the first language of finance and economics

## Scope of Math Concepts

**Operational and Fractional Reasoning** - around 5 - 10

Basic Math Operations, multiplicative reasoning, decimals & fractions

**Proportional Reasoning** - around 10 - 14

RATES, RATIOS & GRAPHING SIMPLE RELATIONSHIPS IN THE QUADRATIC PLAN

**Algebraic Reasoning** - around 11 - 16

SOLVING & GRAPHING SIMPLE FUNCTIONS & EQUATIONS

**Calculus (Marginal Reasoning)** - around 16 - 20

SOLVING COMPLEX FUNCTIONS, DIFFERENTIATION & INTEGRATION  
(true marginal or economic thought)

**Statistical Analysis (Data & Applied Reasoning)** - around 18 - 22

Often includes higher-levels of math such as linear algebra; however, given the use of statistical programs this area has become less theoretical and more technical. It is often now centered around the development of big data coding solutions.



# Marginal Analysis

mathematics can be used to measure...

## Marginal benefit vs. Marginal cost

Marginal means "extra" and is related to what has changed over a period of time. Economic's focus on marginality often results in a need to calculate small units of change.

Calculus, and in particular derivatives, are great for this!

However, we will often simplify mathematical expressions, focusing on the use of algebra to support the internationalization of mathematic concepts. Ratios can also be very helpful here!

**Helpful Brain Tip:** Start to connect the word each with per or “one”.

# Economic Modeling

02 | Section



## **One Economic Idea, Multiple Forms**

**Plato's Theory of Forms gives us a useful analogy for economic modeling: the same underlying relationship can appear in different forms. In economics, we often move between three representations of the same information:**

### **1. Tabular Form — Data**

**Rows and columns organize quantities, values, categories, and observations.  
A table lets us identify patterns in economic information numerically and linguistically.**

### **2. Mathematical Form — Relationship**

**Equations use numbers, variables, and mathematical symbols to express how economic quantities relate to one another.  
They allow us to describe a relationship precisely and calculate values that may not be directly shown in the data.**

### **3. Graphical Form — Model**

**A coordinate plane translates the relationship into space and shape.  
Axes represent variables, while points, lines, slopes, and curves help us visualize direction, magnitude, and change.  
Then I would put one line across the bottom of the slide:  
Table → Equation → Graph: different representations, same economic relationship.**

# model a consumer

04



Part



# Budget Constraint

## Limited Income | Unlimited Desires

For each economic concepts there are often three mathematical forms that help structure our thought and support our understanding of the concept.

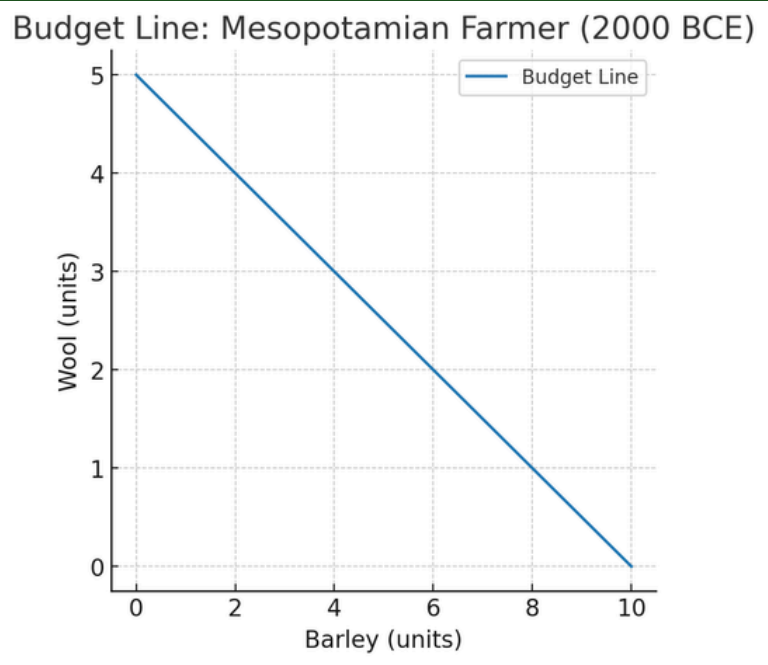
- 1. Table Form
- 2. Expression Form
- 3. Visual Form (Often a Graph)

| Barley (units) | Wool (units) |
|----------------|--------------|
| 0              | 5            |
| 2              | 4            |
| 4              | 3            |
| 6              | 2            |
| 8              | 1            |
| 10             | 0            |

$$p_x(X) + p_y(Y) = M$$

$$2(\text{barley}) + 4(\text{wool}) = 20$$

$$2(\text{barley}) + 4(\text{wool}) = 20$$

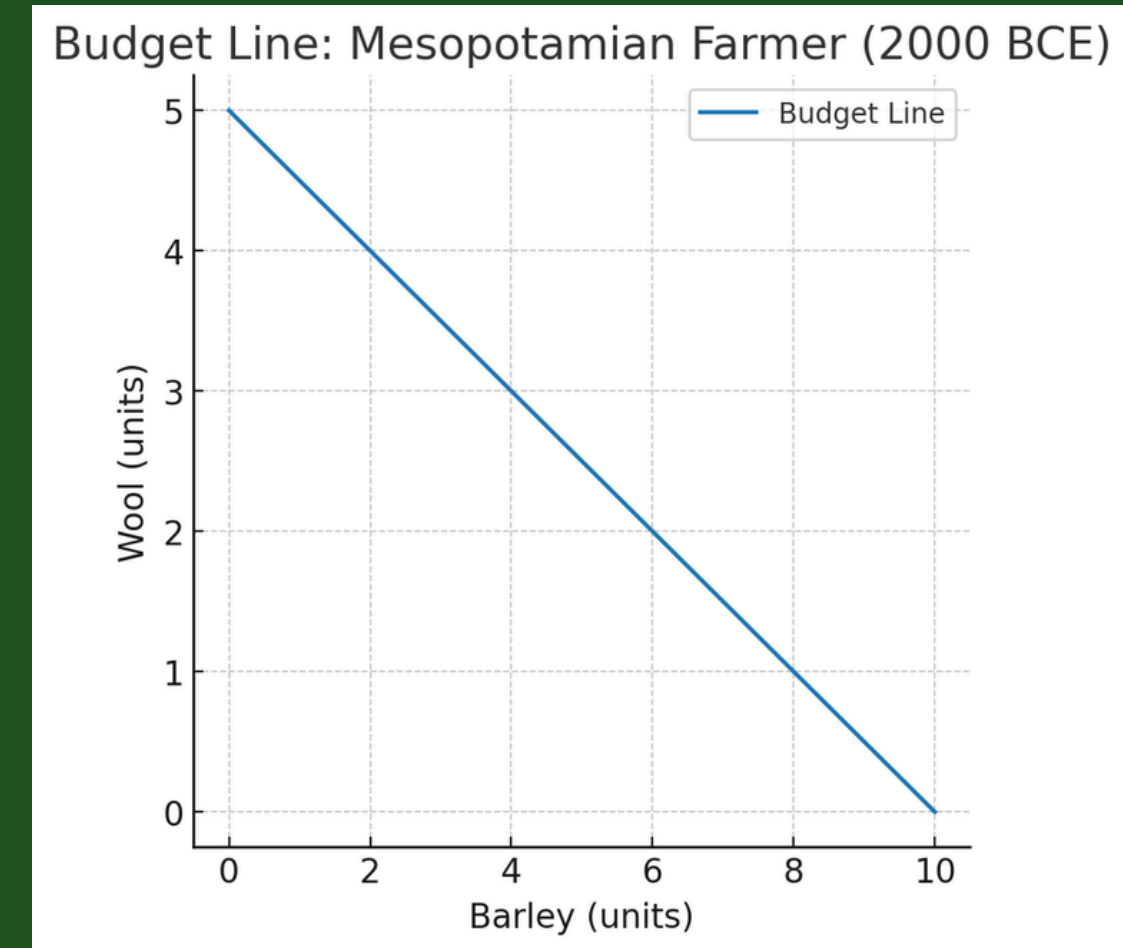


# Budget Constraint

## Limited Income | Unlimited Desires

### Rules of Thumb:

1. Below the line - Feasible or Attainable
2. Above the line - Not Feasible or Unattainable
3. Slope reveals trade off for both goods -  $Y/X$
4. Shift the line to the right - increase income
5. Shift the line to the left - decrease income



| Barley (units) | Wool (units) |
|----------------|--------------|
| 0              | 5            |
| 2              | 4            |
| 4              | 3            |
| 6              | 2            |
| 8              | 1            |
| 10             | 0            |

**Opportunity Cost: What you give up**

**Trade-Off: For every 1 unit of wool the farmer must GIVE UP 2 units of barley.**

**In this example, the opportunity cost is 2 units of barley.**

**Opportunity cost is always in terms of the other good!**

# model a producer

05



Part



# Production Possibilities Model

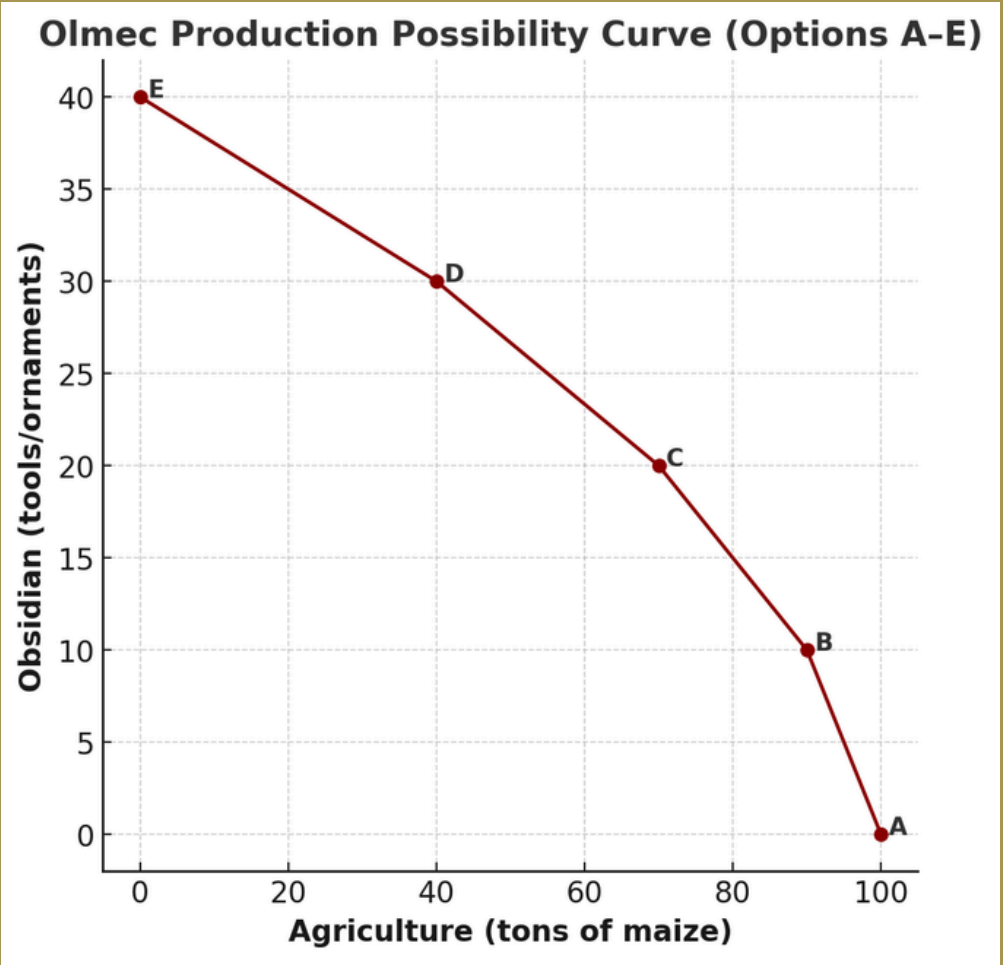
Limited Resources | Limited Technology | Unlimited Desires

**Leading Question:** How does a producer with limited resources and fixed technology make a choice between producing two goods?

**Conceptual Narrative:** If a producer has limited labor and technology, and they are making a choice between two goods, the experience a downward-sloping, curved PPC. This is because as they want more of a good the opportunity cost (expressed in terms of the other good) is increasing, causing an increase slope on our line.

**Law of Increasing Opportunity Cost:** The more you want of good x the more and more you will have to give up good y. The more you want of good y the more and more you will have to give up of good x.

**Allegorical:** The mount to climb gets steeper so you must give up more weight.

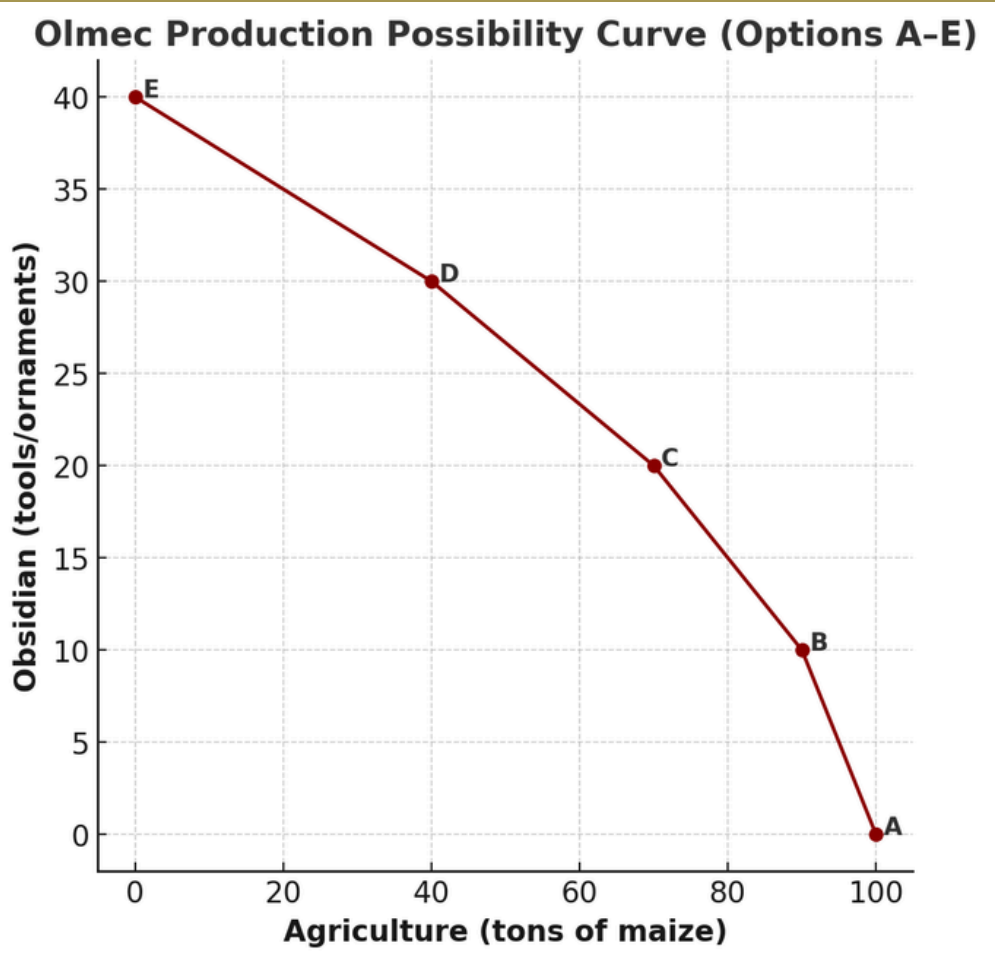


| Option | Agriculture (tons of maize) | Obsidian (tools/ornaments) |
|--------|-----------------------------|----------------------------|
| A      | 100                         | 0                          |
| B      | 90                          | 10                         |
| C      | 70                          | 20                         |
| D      | 40                          | 30                         |
| E      | 0                           | 40                         |

# Production Possibilities Model

Limited Resources | Limited Technology | Unlimited Desires

| Option | Agriculture (tons of maize) | Obsidian (tools/ornaments) |
|--------|-----------------------------|----------------------------|
| A      | 100                         | 0                          |
| B      | 90                          | 10                         |
| C      | 70                          | 20                         |
| D      | 40                          | 30                         |
| E      | 0                           | 40                         |



- Slope:  $Y/X$
- Opportunity Cost: what you give up
- Look to the margins of the table: what math is between the lines?
- Law of Increasing Opportunity Cost: The more you want of good x the more and more you will have to give up good y. The more you want of good y the more and more you will have to give up of good x.
- Allegorical: The mount to climb gets steeper so you must give up more weight.

# model two producers

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# Comparative Advantage & Trade

**Limited Resources** | **Limited Technology** | **Unlimited Desires**

## Leading Question:

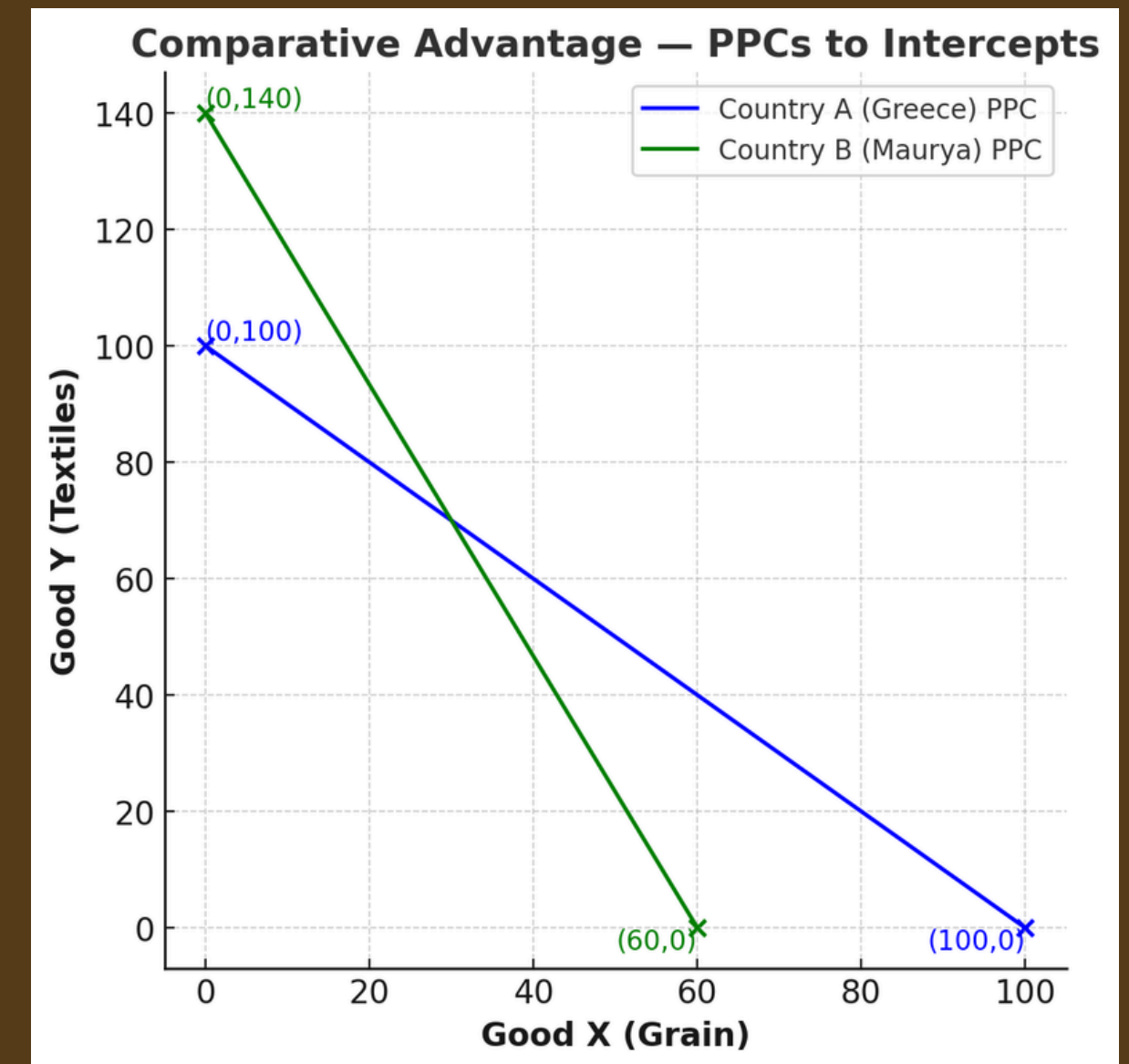
How does a producer make a choice between producing two goods when they have the option to specialize in one good and trade the other?

## Conceptual Narrative:

If two producers are producing a good and there is the possibility for them to trade – we must look to their opportunity costs to understand who should produce what. The key is that the producer with the lower OPPORTUNITY COST – not just the one who makes the most – should specialize in the good and trade for the other.

## Decision-Making:

Focus your work on the goods that require you to give up the least and trade for the other. This will create an overall GAIN FROM TRADE, where output has increased globally.



**Graphing Use:** look at their slope or opportunity cost to determine

**Comparative Advantage** - who gives up the least resources to produce the same quantity (based on OC or slope)

**Absolute Advantage** - who produces the highest quantity given fixed resources

# Independent Practice: Khan Academy

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## **MODULE 1 CONTENT**

Basic Economic Concepts Quiz 1 - REACH MASTERY - we have talked about some of this but most is review of the book.  
Basic Economic Concepts Quiz 2 - REACH MASTERY - we have talked about some of this but most is review of the book.  
Basic Economic Concepts Quiz 3 - Focus on COST-BENEFIT ANALYSIS - Exposure is all that is required.

**HEADS UP:** Utility is covered by Quiz 3 but will not be covered in class until later! Give the questions a try and use this as EXPOSURE to the idea.

## **Homework Rule of Thumb**

If you scored above 80% MOVE ON to the next assignment.

If you scored lower than 80%, retry the assignment 1 to 2 times or until you reach an 80% (take your time)

If you consistently score below 80% on independent practice REACH OUT ASAP AND ATTEND OFFICE HOURS.

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